Repetition (loops)

CSE 101 Section 03
Introduction to Algorithms and Computing
Fall 2022
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Repetition

Display your name 5 times
Display your name 50 times
Display your name 5000002000 times

Repeat the code?

- Bulky code
- Mistakes
- Changes needed in many places
- Monotonous
- Hard to read

2 Types of Repetition Structures

Clap 6 times

- Count controlled
- "For" loop – "*for some number of times do:*"

Clap until I turn around

- Condition Controlled
- "While" loop – "*while this condition is true do:*"

The while Loop: a Condition-Controlled Loop

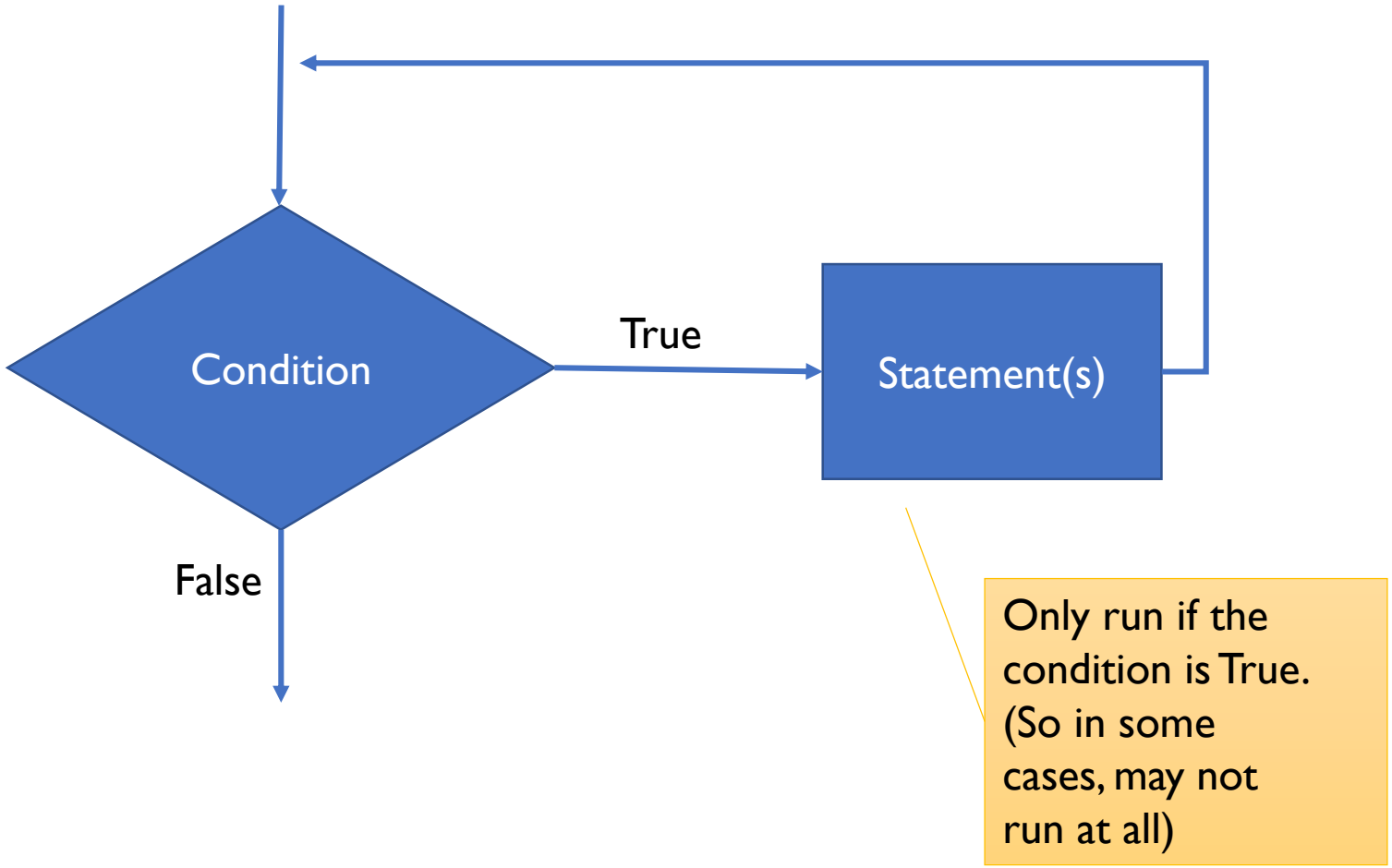
while condition is true, do something

- Two parts:
  - Condition tested for true or false value
  - Statements repeated as long as condition is true

- General format:

```
while condition
        statement
        statement
        statement
end
program continues
```

The while Loop



The while Loop

In order for a loop to stop executing, something has to happen inside the loop to make the condition false.

```
choice = 3
while choice < 5
        disp("Enter 5 to quit")
        choice = input("new choice? ")
end
```

A Matlab function using to get input from the user. Try it!

The while Loop

If there is no change in state to make the loop stop, we have an infinite loop. It will run technically forever.

```
choice = 3
while choice < 5
        disp("Enter 5 to quit")
end
```

The while Loop

- Iteration: one execution of the body of a loop

- while loop is known as a *pretest* loop
  - Tests condition before performing an iteration
    - Will never execute if condition is false to start with
    - Requires performing some steps prior to the loop

Example while Loop

Count backwards from 30 by 7's until you reach zero or a negative number

```
currentNum = 30;
while currentNum > 0
        disp(currentNum)
        currentNum = currentNum - 7;
end
disp("Finished!!")
```

Count-Controlled Loops

- Do 30 push-ups
- Clap your hands 12 times
- Print the happy birthday song 15 times

The for Loop: a Count-Controlled Loop

iterates a specific number of times

General format:

```
for variable = values
        statement(s)
end
```

- Target variable: the variable which is the target of the assignment at the beginning of each iteration

- You should NOT modify the target variable inside the loop – this is done automatically!

The for Loop

- 1st iteration

```
for num = [1 2 3 4 5]
        disp(num)
end
```

- 2nd iteration

```
for num = [1 2 3 4 5]
        disp(num)
end
```

- 3rd iteration

```
for num = [1 2 3 4 5]
        disp(num)
end
```

- 4th iteration

```
for num = [1 2 3 4 5]
        disp(num)
end
```

- 5th iteration

```
for num = [1 2 3 4 5]
        disp(num)
end
```

The for Loop

There are a few ways to have Matlab generate the target variable values

- The colon operator **:**
  - Syntax variable = start : finish Increments by one by default
  - Using `i = 1:5` will generate the values 1 ,2 ,3 ,4 ,5 for `i`
- You can also have different increments :
  - Syntax variable = start : step size : finish
  - Using `i = 1:2:10` will generate the values 1, 3, 5, 7, 9 for `i`
- Can also decrement with negative step
  - Using `i = 10:-2:1` will generate the values 10, 8, 6, 4, 2 for `i`
- You can also provide a vector of values :
  - Using `i = [5 2 9]` will generate the values 5, 2, 9 for `i`

Examples for Loop

```
for i = 1:2:10
    disp(i);
end
```

MATLAB OUTPUT:

1
3
5
7
9

```
for i = 10:-2:1
    disp(i);
end
```

MATLAB OUTPUT:

10
8
6
4
2

Examples for Loop

- Calculate square of each even number less than or equal to10

```
for i = 0:2:10
        disp(i^2)
end
```

NOTE – we are NOT changing the value of i, just using the value.

Next…

Practice Exercises: Selection and Repetition